



DATABASE & CLOUD SYSTEM (BERR2243)

WEEK 5 EXERCISE

Group H

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COURSE	BERR 3 S5
LECTURE'S NAME	DR SOO

Database Exercise WEEK 5

Part 1 Identify Entities & Relationships

Task 1 List core entities

User

- Customers
- Drivers
- Admins

Ride

- Trip event
- Location details

Vehicle

- Car details that linked to specific driver

Payment

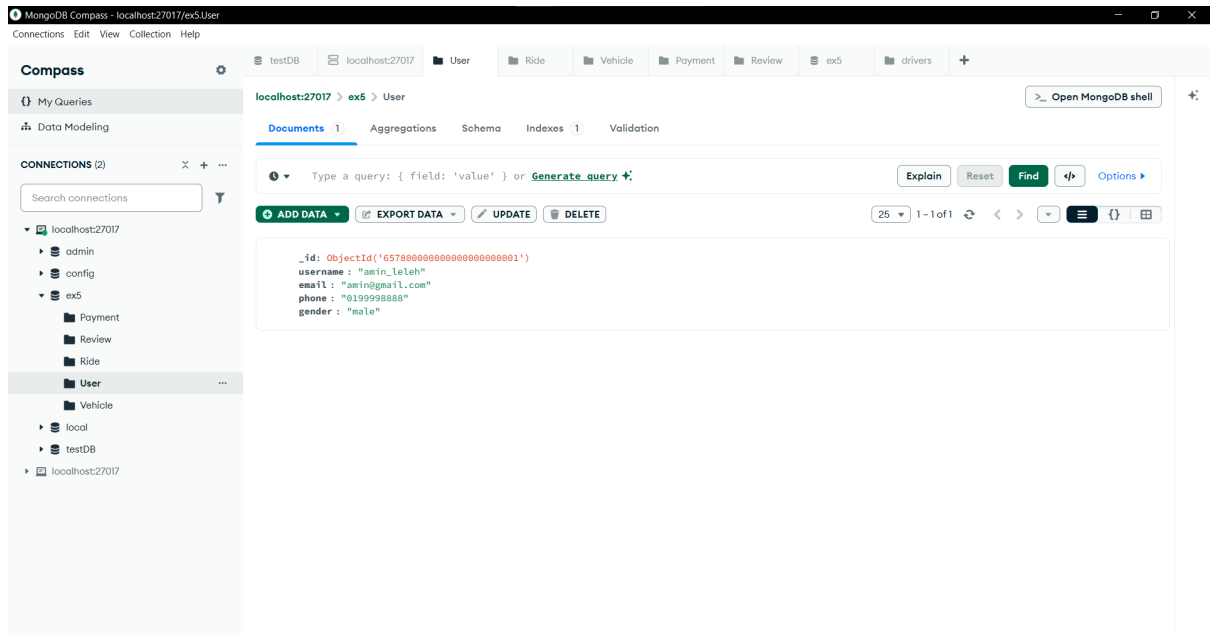
- Payment method
- Amount
- Payment status

Review

- Store the rating
- Feedback comments left by the customer for driver

Task 2 Define attributes

User



MongoDB Compass - localhost:27017/ex5.User

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

localhost:27017 > ex5 > User

Documents 1 Aggregations Schema Indexes 1 Validation

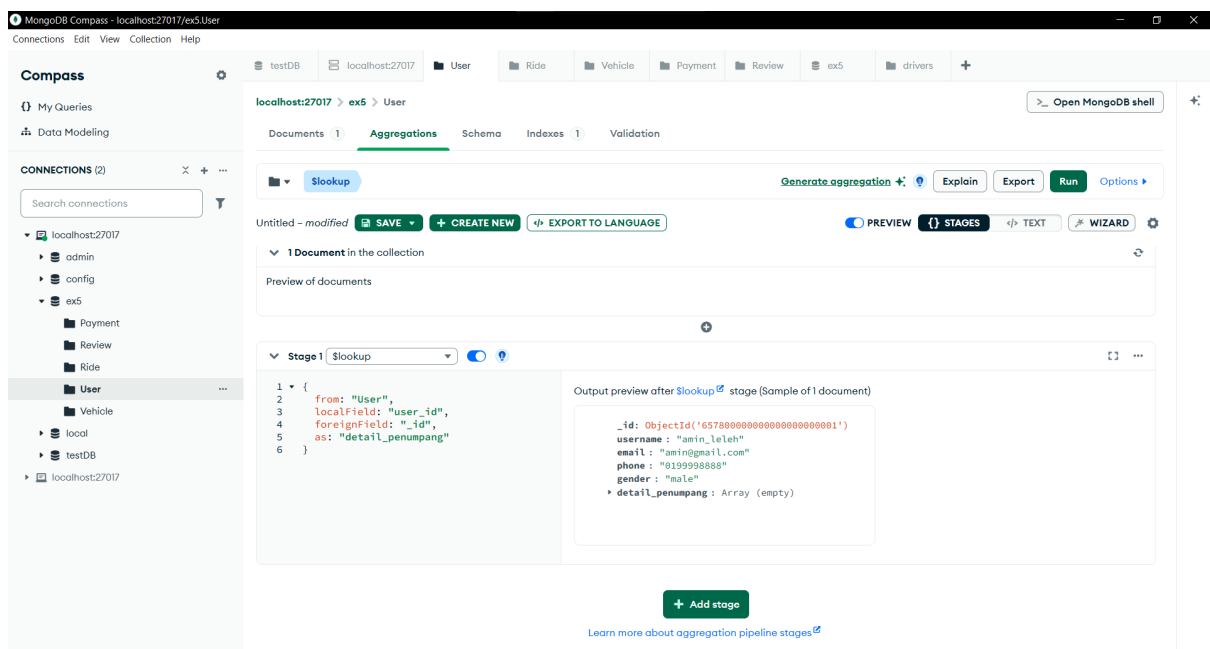
Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find </> Options

ADD DATA EXPORT DATA UPDATE DELETE

25 1-1 of 1

```
{
  "_id": ObjectId('65780000000000000000000000000001'),
  "username": "amin_leleh",
  "email": "amin@gmail.com",
  "phone": "0199998888",
  "gender": "male"
}
```



MongoDB Compass - localhost:27017/ex5.User

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

localhost:27017 > ex5 > User

Documents 1 Aggregations Schema Indexes 1 Validation

[Generate aggregation](#) Explain Export Run Options

Untitled - modified SAVE CREATE NEW EXPORT TO LANGUAGE

PREVIEW STAGES TEXT WIZARD

1 Document in the collection

Preview of documents

Stage 1 \$lookup

```
1 {
2   $lookup: {
3     from: "User",
4     localField: "user_id",
5     foreignField: "_id",
6     as: "detail_penumpang"
7   }
8 }
```

Output preview after \$lookup stage (Sample of 1 document)

```
{
  "_id": ObjectId('65780000000000000000000000000001'),
  "username": "amin_leleh",
  "email": "amin@gmail.com",
  "phone": "0199998888",
  "gender": "male",
  "detail_penumpang": Array (empty)
}
```

+ Add stage

[Learn more about aggregation pipeline stages](#)

Ride

MongoDB Compass - localhost:27017/ex5.Ride

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

- admin
- config
- ex5
 - Payment
 - Review
 - Ride
 - User
 - Vehicle
- local
- testDB

localhost:27017

localhost:27017 > ex5 > Ride

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE

25 1 - 1 of 1

```
{
  "_id": ObjectId('657800000000000000000003'),
  "user_id": ObjectId('657800000000000000000001'),
  "vehicle_id": ObjectId('657800000000000000000002'),
  "pickup": "Kolej Kediaman A",
  "dropoff": "Aeon Bandaraya",
  "fare": 12.5,
  "status": "completed"
}
```

MongoDB Compass - localhost:27017/ex5.Ride

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

- admin
- config
- ex5
 - Payment
 - Review
 - Ride
 - User
 - Vehicle
- local
- testDB

localhost:27017

localhost:27017 > ex5 > Ride

Documents 1 Aggregations Schema Indexes 1 Validation

[Lookup](#) [Generate aggregation](#) [Explain](#) [Export](#) [Run](#) [Options](#)

Untitled - modified [SAVE](#) [+ CREATE NEW](#) [EXPORT TO LANGUAGE](#) [PREVIEW](#) [STAGES](#) [TEXT](#) [WIZARD](#)

1 Document in the collection

Preview of documents

Stage 1 \$lookup

```
1 {
2   from: "User",
3   localField: "user_id",
4   foreignField: "_id",
5   as: "detail_penumpang"
6 }
```

Output preview after \$lookup stage (Sample of 1 document)

```
{
  "_id": ObjectId('657800000000000000000003'),
  "user_id": ObjectId('657800000000000000000001'),
  "vehicle_id": ObjectId('657800000000000000000002'),
  "pickup": "Kolej Kediaman A",
  "dropoff": "Aeon Bandaraya",
  "fare": 12.5,
  "status": "completed",
  "detail_penumpang": Array (1)
}
```

[+ Add stage](#)

[Learn more about aggregation pipeline stages](#)

Vehicle

MongoDB Compass - localhost:27017/ex5.Vehicle

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

- admin
- config
- ex5
 - Payment
 - Review
 - Ride
 - User
 - Vehicle
- local
- testDB
- localhost:27017

localhost:27017 > ex5 > Vehicle

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE

25 1-1 of 1

```
{
  "_id": ObjectId('65780000000000000000000000000002'),
  "plate_number": "JEV 8888",
  "model": "Honda Civic",
  "color": "Putih Mutiara",
  "driver_name": "Abang Jago",
  "status": "available"
}
```

MongoDB Compass - localhost:27017/ex5.Vehicle

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

- admin
- config
- ex5
 - Payment
 - Review
 - Ride
 - User
 - Vehicle
- local
- testDB
- localhost:27017

localhost:27017 > ex5 > Vehicle

Documents 1 Aggregations Schema Indexes 1 Validation

Generate aggregation

Explain Export Run Options

Untitled - modified SAVE CREATE NEW EXPORT TO LANGUAGE

PREVIEW STAGES TEXT WIZARD

1 Document in the collection

Preview of documents

Stage 1 \$lookup

```
1 {
2   from: "Vehicle",
3   localField: "vehicle_id",
4   foreignField: "_id",
5   as: "detail_kereta"
6 }
```

Output preview after \$lookup stage (Sample of 1 document)

```
{
  "_id": ObjectId('65780000000000000000000000000002'),
  "plate_number": "JEV 8888",
  "model": "Honda Civic",
  "color": "Putih Mutiara",
  "driver_name": "Abang Jago",
  "status": "available",
  "detail_kereta": Array (empty)
}
```

+ Add stage

[Learn more about aggregation pipeline stages](#)

Payment

MongoDB Compass - localhost:27017/ex5.Payment

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

- admin
- config
- ex5
 - Payment
 - Review
 - Ride
 - User
 - Vehicle
- local
- testDB

localhost:27017

localhost:27017 > ex5 > Payment

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE

25 1 - 1 of 1

```
{
  "_id": ObjectId('657800000000000000000004'),
  "ride_id": ObjectId('657800000000000000000003'),
  "amount": 12.5,
  "method": "Cash",
  "transaction_date": 2023-12-12T14:00:00.000+00:00
}
```

MongoDB Compass - localhost:27017/ex5.Payment

Connections Edit View Collection Help

testDB localhost:27017 User Ride Vehicle Payment Review ex5 drivers +

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

- admin
- config
- ex5
 - Payment
 - Review
 - Ride
 - User
 - Vehicle
- local
- testDB

localhost:27017

localhost:27017 > ex5 > Payment

Documents 1 Aggregations Schema Indexes 1 Validation

[Lookup](#) [Generate aggregation](#) Explain Export Run Options

Untitled - modified [SAVE](#) [+ CREATE NEW](#) [EXPORT TO LANGUAGE](#) [PREVIEW](#) [STAGES](#) [TEXT](#) [WIZARD](#)

1 Document in the collection

Preview of documents

Stage 1 [Lookup](#) [ON](#) [?](#)

```
1 {
2   from: "Payment",
3   localField: "_id",
4   foreignField: "ride_id",
5   as: "info_bayaran"
6 }
```

Output preview after [Lookup](#) stage (Sample of 1 document)

```
{
  "_id": ObjectId('657800000000000000000004'),
  "ride_id": ObjectId('657800000000000000000003'),
  "amount": 12.5,
  "method": "Cash",
  "transaction_date": 2023-12-12T14:00:00.000+00:00,
  "info_bayaran": Array (empty)
}
```

[+ Add stage](#)

[Learn more about aggregation pipeline stages](#)

Review

MongoDB Compass - localhost:27017/ex5/Review

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

admin

config

ex5

Payment

Review

Ride

User

Vehicle

local

testDB

localhost:27017

testDB

localhost:27017

User

Ride

Vehicle

Payment

Review

ex5

drivers

+

localhost:27017 > ex5 > Review

>_ Open MongoDB shell

Documents 1

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

25

1-1 of 1

<

>

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```
_id: ObjectId('657800000000000000000005')
ride_id: ObjectId('657800000000000000000003')
user_id: ObjectId('657800000000000000000001')
rating: 5
comment: "Mantap kereta wangi, driver laju!"
created_at: 2023-12-12T14:30:00.000+00:00
```

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the connected databases: 'localhost:27017' and 'testDB'. The 'Review' collection is selected. The main panel displays the 'Aggregations' tab for the 'Review' collection. The aggregation pipeline consists of a single stage, '\$lookup', which is configured to join the 'Review' collection with the 'localField: '_id'', 'foreignField: 'ride_id'', and 'as: 'bintang_review'' fields. The output preview shows a sample document with the joined data.

```

1 {
2   $lookup: {
3     from: "Review",
4     localField: "_id",
5     foreignField: "ride_id",
6     as: "bintang_review"
7   }
8 }

```

Output preview after \$lookup stage (Sample of 1 document)

```

{
  _id: ObjectId('657800000000000000000000'),
  ride_id: ObjectId('657800000000000000000000'),
  user_id: ObjectId('657800000000000000000000'),
  rating: 5,
  comment: "Mantap kereta wangi, driver laju!",
  created_at: 2023-12-12T14:30:00.000+00:00,
  bintang_review: Array (empty)
}

```

At the bottom, there is a '+ Add stage' button and a link to 'Learn more about aggregation pipeline stages'.

Task 3 Model relationship (decide how entities relate)

- User > Ride: One-to-Many (1 User can have many rides over time)
- Driver > Vehicle: One-to-One (1 Driver has 1 active Vehicle)
- Ride > Payment: One-to-One (1 Ride generates exactly 1 Payment transaction)
- Ride > Review: One-to-One (1 Ride can have 1 Review)

Part 2 Design the ERD

Task 4 Draw the ERD

